GEOMETRY · UNIT 8: TRANSFORMATIONS

Animation Studio

Animate a game sprite with translations, reflections, rotations, and dilations.

Honors

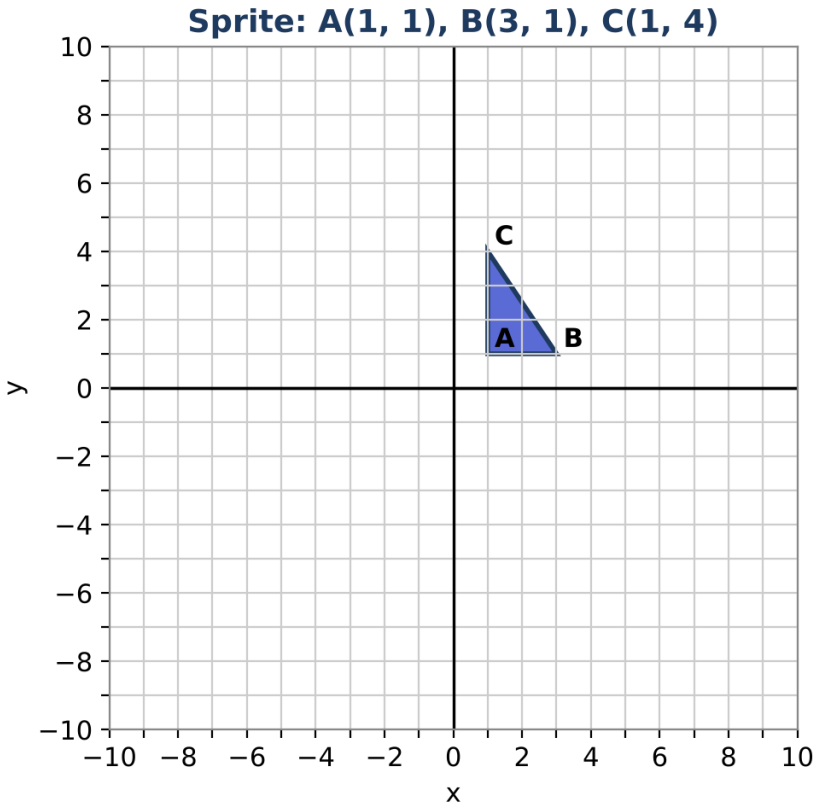
2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

A small game studio needs an animator. Their sprite is a triangle with vertices A(1, 1), B(3, 1), and C(1, 4), and every move in the game is a transformation.

Game engines move sprites with coordinate rules, so you will write and apply rules, combine moves, and storyboard a short animation.



Learning Goals

- I can rotate and dilate about any center.
- I can find the line of reflection that maps a figure to its image.
- I can prove facts about compositions of reflections.

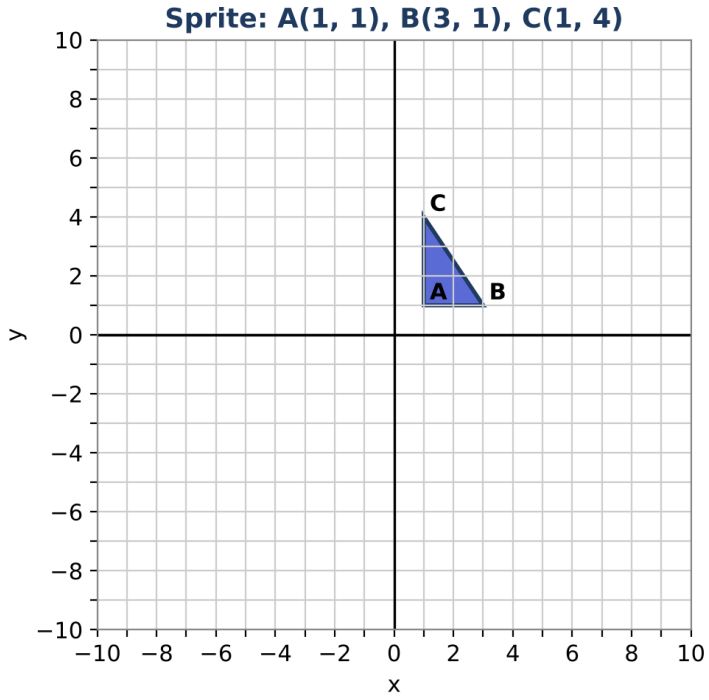
Project Timeline

When	What you will do
Day 1	Moves About Any Point
Day 2	Reflections & Compositions
Day 3	Storyboard with Proof
Final	Animation Storyboard + reflection

Part 1: Moves About Any Point

Day 1 · Center the move at a vertex of the sprite.

1. Rotate the sprite 90° counterclockwise about A(1, 1).



2. Dilate the sprite by 3 with center A(1, 1). Is A′ different from A?

3. Write a general rule for rotating 90° counterclockwise about (h, k).

Part 2: Reflections & Compositions

Day 2 · Justify each answer.

4. A reflection maps A(1, 1) to (5, 3). Find the equation of the line of reflection.

5. Reflect the sprite over x = 2 and then over x = 5. Show that the result is a translation and find its distance.

6. Reflect over the x-axis and then over y = x. Show that the result is a rotation and find its angle.

Part 3: Storyboard with Proof

Day 3 · Design a looping animation of at least six frames that uses a rotation about a non-origin point.

7. Plan the frames and rules.

8. List all coordinates.

9. Prove that your loop returns to the start using compositions.

Final Product: Animation Storyboard

Your final product must include:

- Non-origin rotation and dilation
- Line of reflection
- Composition proofs
- Looping storyboard with proof

Reflection

R1. How are two reflections related to a translation or a rotation?

R2. Why does the center of a dilation stay fixed?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Animation uses at least three different transformations with correct rules and coordinates. Goes beyond the minimum with a creative, realistic choice.	Animation uses at least three different transformations with correct rules and coordinates.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Storyboard shows each frame on a grid with the coordinate rule and the image coordinates. Polished, organized, and visually engaging.	Storyboard shows each frame on a grid with the coordinate rule and the image coordinates. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.